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Paper title: Review on electromyography signal acquisition and processing

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- **Research field of the orientative paper:** This paper reviews biomedical signal processing, specifically focusing on electromyography (EMG) technology for recording electrical signals from neuromuscular activities. Applications span medical diagnostics, prosthetic control, speech recognition, and biomechanical analysis.

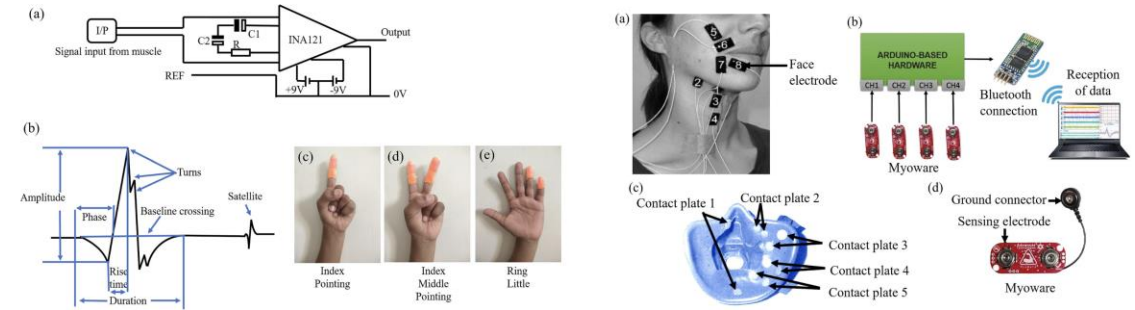
- **Make a summary of the paper:** The authors provide a comprehensive review of EMG signal acquisition methods and processing techniques. They compare surface electrodes versus needle electrodes, analyze recognition accuracy across various applications (robotics, speech recognition, diagnostics), and identify that while signal recognition accuracy exceeds 99%, capture efficiency remains limited at ~70%, indicating significant room for hardware improvement.

- **Explain the contribution of the paper in respect other papers (state of the art). Perhaps it will be useful to fill in this section once make the global discussion of all the papers.**

In this paper there are some novel aspects, like as a systematic comparison of capture efficiency vs. recognition accuracy across multiple studies:

- Practical design guidelines for EMG system development (8 specific recommendations)
- Identification of hardware bottleneck: Recognition algorithms are mature (>99%), but sensors need improvement
- Gap addressed: Previous reviews focused on algorithms; this paper emphasizes the hardware limitation as the primary barrier to EMG technology advancement.

- **Explain the results presented in the paper. Use, if necessary, graphs or plots of the paper:** In this review paper, we explain how the different types of EMG signals are acquired and processed. This paper will be useful for medical and engineering communities for developing better diagnostics using EMG. Figure 2 shows the basic temporal characteristics of the EMG signal and Figure 3 shows Different positions on the face, where electrodes should be placed to acquire proper EMG signal.



- **Critical reasoning about the results explained in the paper:**

The statement "recognition accuracy has already crossed 99%" is misleading, this applies only to specific tasks (MUAP decomposition), not general EMG classification where 68-93% is more typical.

The proposed solutions (high-density EMG, single-cell detection, open databases) are reasonable but lack concrete implementation strategies or timeline feasibility analysis. The noise reduction recommendations are particularly vague ("keep cables short" is overly simplistic for a 2020 review).